

Homework 08: Chapter 10

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (10 points, Python)

Create a class `Complex` which implements complex number arithmetic:

1. The class should have two attributes:

- `_real`
- `_imag`

2. The class should have these magic methods:

- `__str__` should return a string in the following format: `a+bi`
- `__add__` should return a new `Complex` object that is the sum of the two operands
- `__sub__` should return a new `Complex` object that is the difference of the two operands
- `__mul__` should return a new `Complex` object that is the product of the two operands
- `__truediv__` should return a new `Complex` object that is the quotient of the two operands

3. The class should have one additional method:

- `conj` (complex conjugate)

Here's some examples the class and methods:

For reference, here are some definitions for complex arithmetic. Let $x = a + bi$ and $y = c + di$ be two complex numbers.

The sum of x and y is:

$$\begin{aligned}x + y &= (a + bi) + (c + di) \\ &= (a + c) + (b + d)i\end{aligned}$$

The difference of x and y is:

$$\begin{aligned}x - y &= (a + bi) - (c + di) \\ &= (a - c) + (b - d)i\end{aligned}$$

The product of x and y is:

$$\begin{aligned}xy &= (a + bi)(c + di) \\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

The quotient of x and y is:

$$\begin{aligned}\frac{x}{y} &= \frac{a + bi}{c + di} \\ &= \frac{(ac + bd)}{c^2 + d^2} + \frac{(bc - ad)}{c^2 + d^2}i\end{aligned}$$

Exercise 2 (20 points, Python)

1. Write two functions that perform the same task in different ways. The task should take anywhere between (roughly) 0.1 and 1 seconds to complete (basically it needs to take a while).
2. Use the `time` module to find the average time it takes to perform each function. Clearly print average times, clearly label each version.